

Scene Generation Benchmark

Genere le 2026-08-02 16:08:36.

Ce document audite la generation directe de `.scene` par `Ministral` a partir de briefs spatiaux fixes.

Pipeline :

- le moteur fabrique le prompt exact via `--dump-scene-audit-prompt`
- `llama-cli` genere une reponse brute
- la reponse est normalisee en `.scene`
- le renderer audite le fichier et produit une image si la scene est valide

Parametres :

- rendu : `800x400`, 8 spp, 3 bounces, 1 direct samples
- LLM : `temperature=0.0`, `n_predict=1024`, `ctx=4096`, `seed=42`

Resultat global : `9/10` scenes valides, `1` invalides.

Case	Status	Gen ms	Audit ms	Triangles	Materials
<code>entry_gate_dusk</code>	<code>valid</code>	18067.30	462.15	220	20
<code>sally_port_checkpoint</code>	<code>valid</code>	20972.08	582.50	518	44
<code>badge_vestibule</code>	<code>valid</code>	21665.24	630.95	276	28
<code>server_aisles_dense</code>	<code>valid</code>	23216.06	612.73	1240	105
<code>control_hub</code>	<code>valid</code>	18988.96	559.02	166	18
<code>backup_vault</code>	<code>valid</code>	24317.32	774.31	772	66
<code>cooling_exchange_bay</code>	<code>valid</code>	26772.40	968.71	736	63
<code>loading_dock_dust</code>	<code>invalid_scene</code>	21240.09	95.15	0	0
<code>roof_watch_dusk</code>	<code>valid</code>	20044.93	577.93	410	35
<code>stairhead_parapet</code>	<code>valid</code>	20270.69	504.36	378	34

Entry Gate

A concrete gate court faces the datacenter threshold. Dust clings to the bars and the badge reader blinks beside a service intercom.

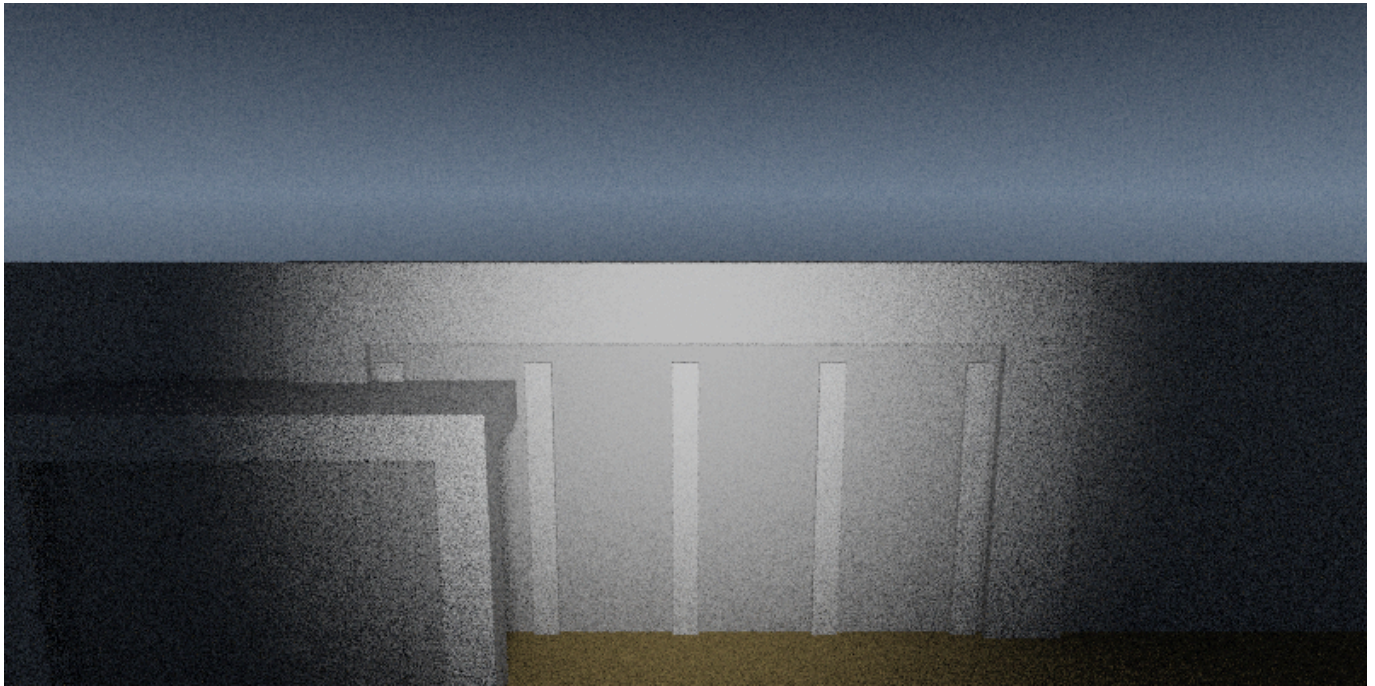
Status : `valid` Slug : `entry_gate_dusk` Generation : `18067.30 ms` Audit + render : `462.15 ms` Scene lines : `11` Scene chars : `884` Triangles : `220` Materials : `20`

State : `generated/scene_generation_benchmark/states/entry_gate_dusk.session.json` Prompt : `generated/scene_generation_benchmark/prompts/entry_gate_dusk.prompt.txt` Raw response : `generated/scene_generation_benchmark/responses/entry_gate_dusk.raw.txt` Scene :

generated/scene_generation_benchmark/scenes/entry_gate_dusk.scene Audit stdout :
generated/scene_generation_benchmark/logs/entry_gate_dusk.audit.stdout.txt Audit stderr :
generated/scene_generation_benchmark/logs/entry_gate_dusk.audit.stderr.txt

Spatial brief :

- location_id: **gate**
- location_archetype: **threshold** courtyard
- time_of_day: **dusk**
- visibility_level: **clear**
- desert_state: **still**
- interior_density: **sparse**
- alert_level: **1**
- anchors: **gate** frame, perimeter wall, badge reader, intercom plate
- visible_objects: badge reader, service intercom, warning placard, padlocked crate
- blocked_exits: **west**, **south**
- spatial_anomalies: (**none**)



Sally Port Checkpoint

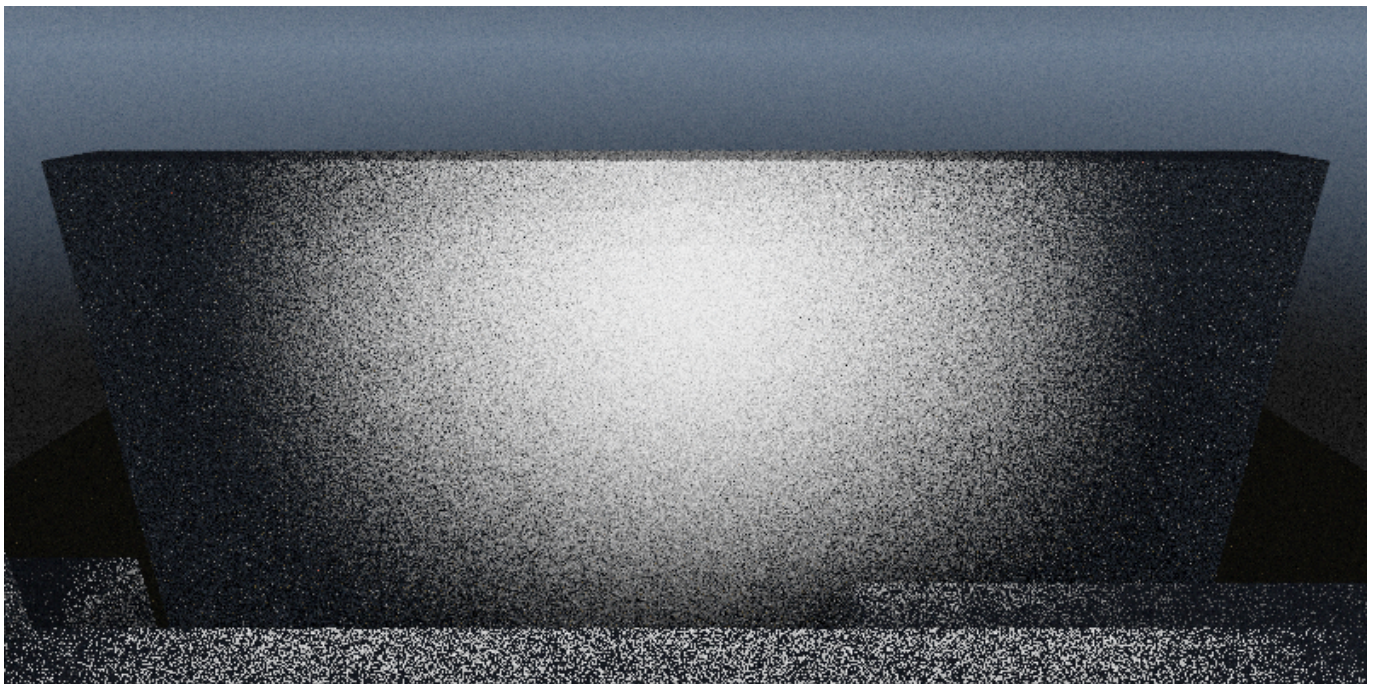
A narrow checkpoint sits between two heavy gates. A code keypad, a dead camera head, and a steel cabinet make the space feel procedural and exposed.

Status : **valid** Slug : **sally_port_checkpoint** Generation : **20972.08 ms** Audit + render : **582.50 ms** Scene lines : **15** Scene chars : **1116** Triangles : **518** Materials : **44**

State : **generated/scene_generation_benchmark/states/sally_port_checkpoint.session.json**
Prompt : **generated/scene_generation_benchmark/prompts/sally_port_checkpoint.prompt.txt** Raw response : **generated/scene_generation_benchmark/responses/sally_port_checkpoint.raw.txt**
Scene : **generated/scene_generation_benchmark/scenes/sally_port_checkpoint.scene** Audit stdout : **generated/scene_generation_benchmark/logs/sally_port_checkpoint.audit.stdout.txt** Audit stderr : **generated/scene_generation_benchmark/logs/sally_port_checkpoint.audit.stderr.txt**

Spatial brief :

- location_id: `unknown`
- location_archetype: `security checkpoint`
- time_of_day: `night`
- visibility_level: `low`
- desert_state: `windy`
- interior_density: `sparse`
- alert_level: `2`
- anchors: `inner gate, outer gate, checkpoint alcove`
- visible_objects: `code keypad, steel cabinet, dead camera head, inspection hatch`
- blocked_exits: `south`
- spatial_anomalies: `camera loop stuck on static`



Badge Vestibule

A bare vestibule buffers the exterior from the datacenter core. The only relief is a reader column, a maintenance hatch, and a wall placard with faded rules.

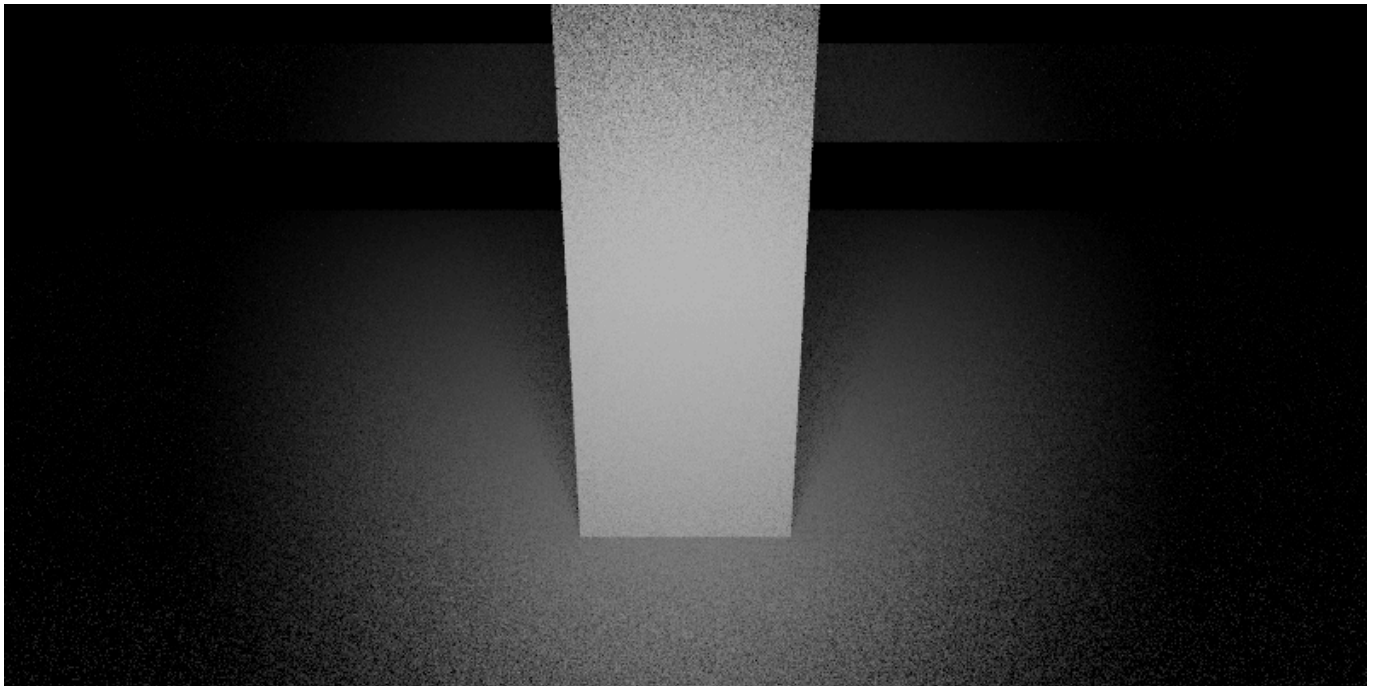
Status : `valid` Slug : `badge_vestibule` Generation : `21665.24 ms` Audit + render : `630.95 ms` Scene lines : `15` Scene chars : `1124` Triangles : `276` Materials : `28`

State : `generated/scene_generation_benchmark/states/badge_vestibule.session.json` Prompt : `generated/scene_generation_benchmark/prompts/badge_vestibule.prompt.txt` Raw response : `generated/scene_generation_benchmark/responses/badge_vestibule.raw.txt` Scene : `generated/scene_generation_benchmark/scenes/badge_vestibule.scene` Audit stdout : `generated/scene_generation_benchmark/logs/badge_vestibule.audit.stdout.txt` Audit stderr : `generated/scene_generation_benchmark/logs/badge_vestibule.audit.stderr.txt`

Spatial brief :

- location_id: `unknown`

- location_archetype: **airlock vestibule**
- time_of_day: **night**
- visibility_level: **clear**
- desert_state: **still**
- interior_density: **sparse**
- alert_level: **1**
- anchors: **reader column, inner door, outer door**
- visible_objects: **badge reader, maintenance hatch, wall placard, ceiling speaker**
- blocked_exits: **west**
- spatial_anomalies: **(none)**



Primary Server Aisles

Dense aisles of racks run into darkness. A service cart, a floor hatch, and a side console break the repetition under red LEDs.

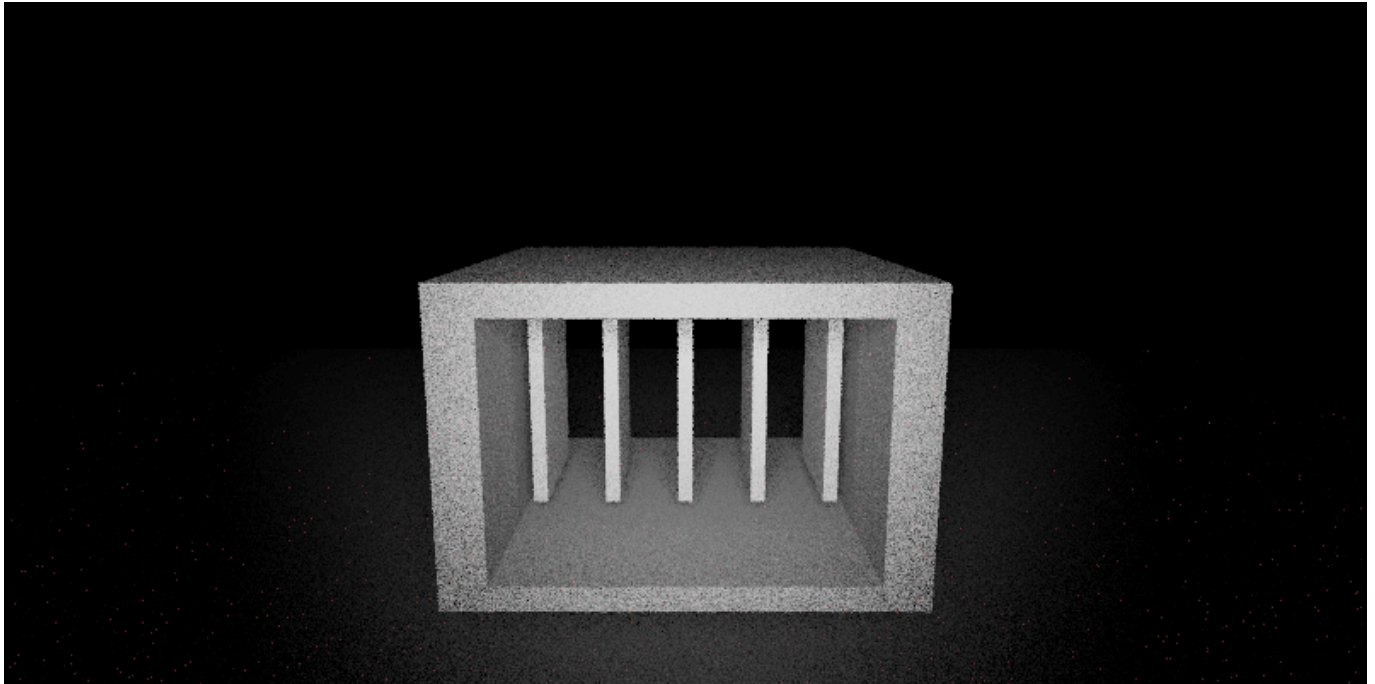
Status : **valid** Slug : **server_aisles_dense** Generation : **23216.06 ms** Audit + render : **612.73 ms** Scene lines : **16** Scene chars : **1251** Triangles : **1240** Materials : **105**

State : **generated/scene_generation_benchmark/states/server_aisles_dense.session.json** Prompt : **generated/scene_generation_benchmark/prompts/server_aisles_dense.prompt.txt** Raw response : **generated/scene_generation_benchmark/responses/server_aisles_dense.raw.txt** Scene : **generated/scene_generation_benchmark/scenes/server_aisles_dense.scene** Audit stdout : **generated/scene_generation_benchmark/logs/server_aisles_dense.audit.stdout.txt** Audit stderr : **generated/scene_generation_benchmark/logs/server_aisles_dense.audit.stderr.txt**

Spatial brief :

- location_id: **server_aisles**
- location_archetype: **server aisles**
- time_of_day: **night**
- visibility_level: **low**

- desert_state: **still**
- interior_density: **dense**
- alert_level: **2**
- anchors: **rack corridor, service lane, ceiling cable tray**
- visible_objects: **service cart, floor hatch, side console, sealed lockbox**
- blocked_exits: **south**
- spatial_anomalies: **cooling hum pulses unevenly**



Sector Control Hub

A compact control room opens off the aisles. One main console dominates the center while a locked cabinet and an access hatch wait at the margins.

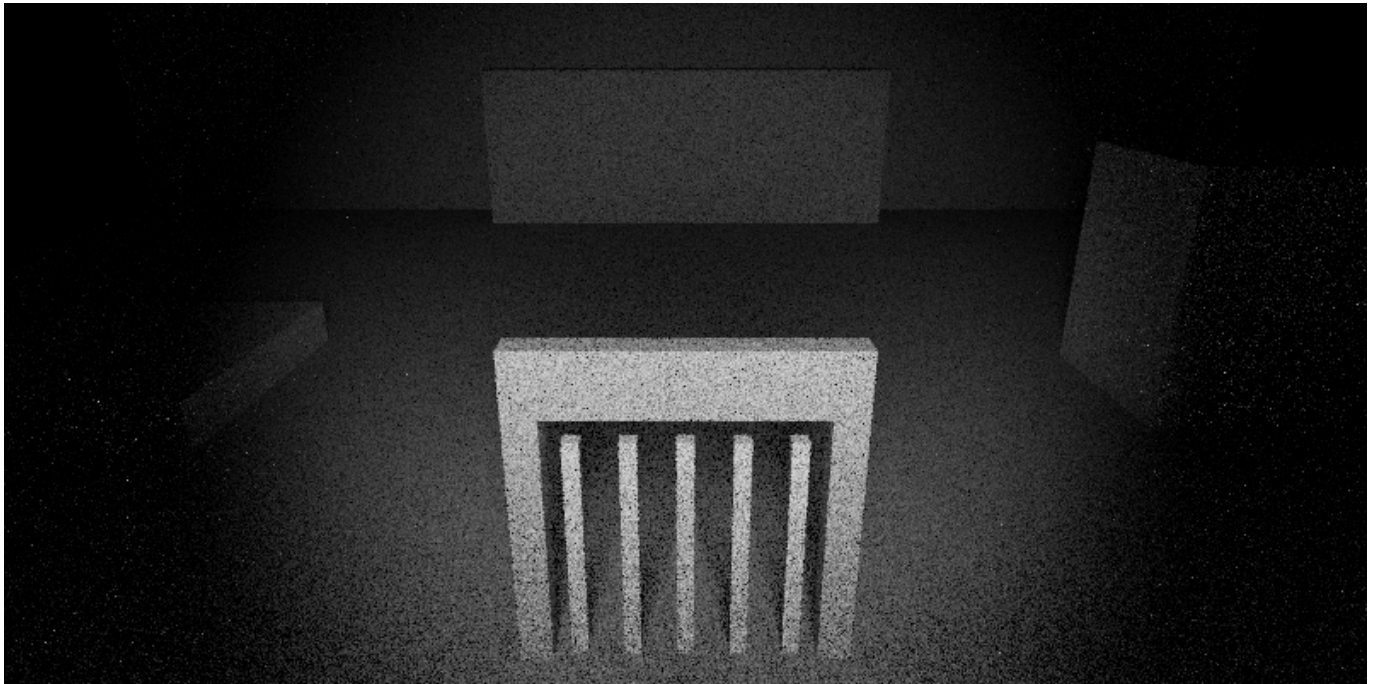
Status : **valid** Slug : **control_hub** Generation : **18988.96 ms** Audit + render : **559.02 ms** Scene lines : **13**
Scene chars : **937** Triangles : **166** Materials : **18**

State : **generated/scene_generation_benchmark/states/control_hub.session.json** Prompt :
generated/scene_generation_benchmark/prompts/control_hub.prompt.txt Raw response :
generated/scene_generation_benchmark/responses/control_hub.raw.txt Scene :
generated/scene_generation_benchmark/scenes/control_hub.scene Audit stdout :
generated/scene_generation_benchmark/logs/control_hub.audit.stdout.txt Audit stderr :
generated/scene_generation_benchmark/logs/control_hub.audit.stderr.txt

Spatial brief :

- location_id: **unknown**
- location_archetype: **control room**
- time_of_day: **night**
- visibility_level: **clear**
- desert_state: **still**
- interior_density: **sparse**
- alert_level: **2**

- anchors: `main console, wall cabinet, floor hatch`
- visible_objects: `main console, locked cabinet, floor hatch, status placard`
- blocked_exits: `east, south`
- spatial_anomalies: `(none)`



Backup Vault

A chilled storage vault holds sealed pods and backup racks. A biometric pad, a wheeled crate, and a suspended service panel suggest expensive access.

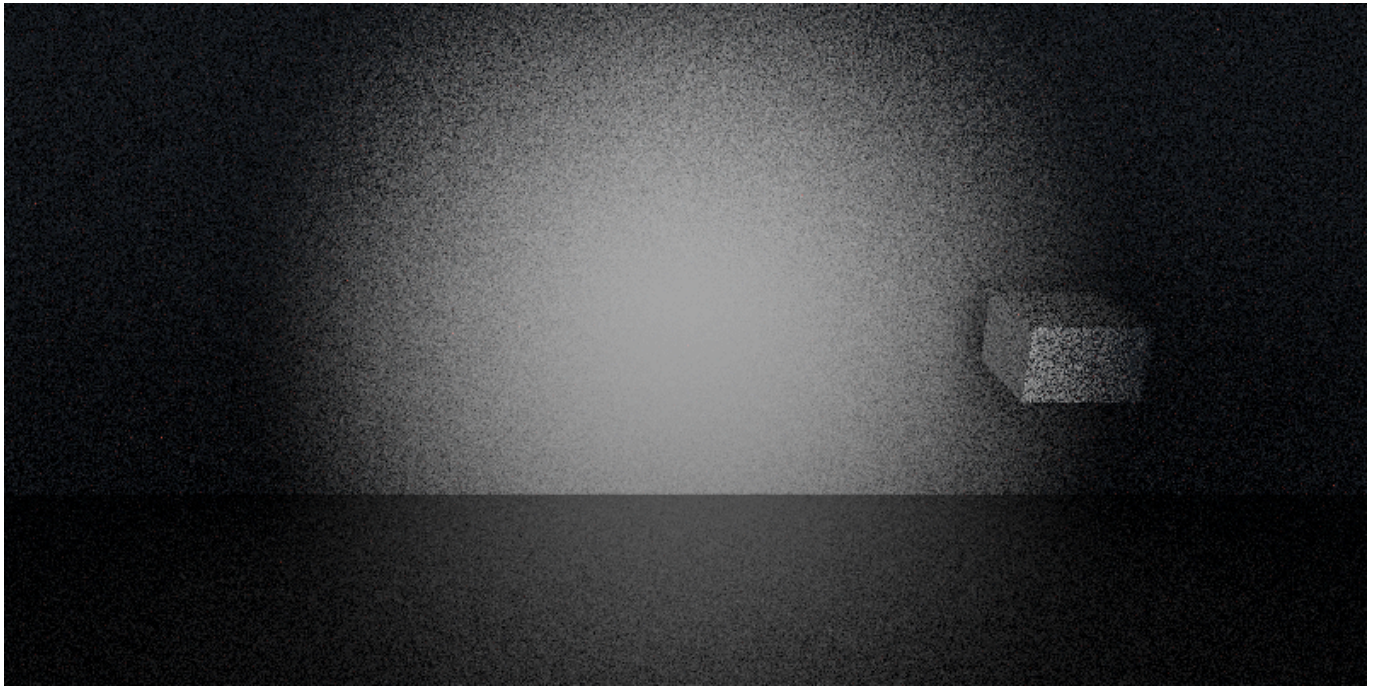
Status : `valid` Slug : `backup_vault` Generation : `24317.32 ms` Audit + render : `774.31 ms` Scene lines : `17`
Scene chars : `1348` Triangles : `772` Materials : `66`

State : `generated/scene_generation_benchmark/states/backup_vault.session.json` Prompt :
`generated/scene_generation_benchmark/prompts/backup_vault.prompt.txt` Raw response :
`generated/scene_generation_benchmark/responses/backup_vault.raw.txt` Scene :
`generated/scene_generation_benchmark/scenes/backup_vault.scene` Audit stdout :
`generated/scene_generation_benchmark/logs/backup_vault.audit.stdout.txt` Audit stderr :
`generated/scene_generation_benchmark/logs/backup_vault.audit.stderr.txt`

Spatial brief :

- location_id: `unknown`
- location_archetype: `backup_vault`
- time_of_day: `night`
- visibility_level: `low`
- desert_state: `still`
- interior_density: `dense`
- alert_level: `3`
- anchors: `sealed data pods, vault door, service panel`
- visible_objects: `biometric pad, wheeled crate, service panel, emergency switch`
- blocked_exits: `north, west`

- spatial_anomalies: cold vapor drifts near the floor



Cooling Exchange Bay

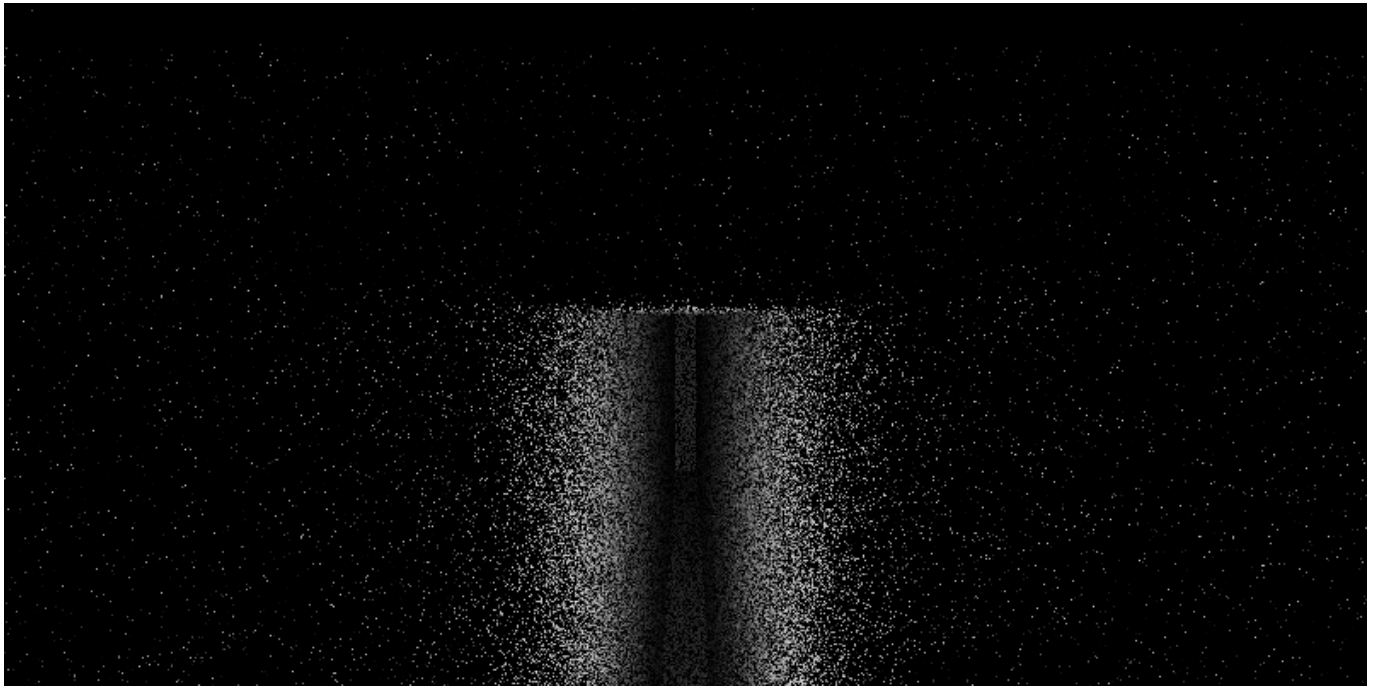
Tall cooling blocks and pipe runs crowd an industrial chamber. A valve wheel, a drip tray, and a maintenance console provide the room's few obvious handles.

Status : **valid** Slug : **cooling_exchange_bay** Generation : **26772.40 ms** Audit + render : **968.71 ms** Scene lines : **19** Scene chars : **1511** Triangles : **736** Materials : **63**

State : **generated/scene_generation_benchmark/states/cooling_exchange_bay.session.json**
Prompt : **generated/scene_generation_benchmark/prompts/cooling_exchange_bay.prompt.txt** Raw response : **generated/scene_generation_benchmark/responses/cooling_exchange_bay.raw.txt**
Scene : **generated/scene_generation_benchmark/scenes/cooling_exchange_bay.scene** Audit stdout : **generated/scene_generation_benchmark/logs/cooling_exchange_bay.audit.stdout.txt** Audit stderr : **generated/scene_generation_benchmark/logs/cooling_exchange_bay.audit.stderr.txt**

Spatial brief :

- location_id: **unknown**
- location_archetype: **cooling plant bay**
- time_of_day: **night**
- visibility_level: **dusty**
- desert_state: **still**
- interior_density: **dense**
- alert_level: **2**
- anchors: **cooling blocks, pipe gallery, drain trench**
- visible_objects: **valve wheel, maintenance console, drip tray, tool locker**
- blocked_exits: **west**
- spatial_anomalies: **one pipe bank rattles under strain**



Dust Loading Dock

A side loading bay opens toward the desert through a half-raised shutter. Pallets, a cargo crate, and a bent lamp post stage a tense threshold between inside and outside.

Status : `invalid_scene` Slug : `loading_dock_dust` Generation : `21240.09 ms` Audit + render : `95.15 ms`
Scene lines : `14` Scene chars : `1103` Triangles : `0` Materials : `0`

State : `generated/scene_generation_benchmark/states/loading_dock_dust.session.json` Prompt :
`generated/scene_generation_benchmark/prompts/loading_dock_dust.prompt.txt` Raw response :
`generated/scene_generation_benchmark/responses/loading_dock_dust.raw.txt` Scene :
`generated/scene_generation_benchmark/scenes/loading_dock_dust.scene` Audit stdout :
`generated/scene_generation_benchmark/logs/loading_dock_dust.audit.stdout.txt` Audit stderr :
`generated/scene_generation_benchmark/logs/loading_dock_dust.audit.stderr.txt`

Spatial brief :

- location_id: `unknown`
- location_archetype: `loading dock`
- time_of_day: `dusk`
- visibility_level: `dusty`
- desert_state: `dusty`
- interior_density: `sparse`
- alert_level: `2`
- anchors: `half-raised shutter, dock lip, desert threshold`
- visible_objects: `cargo crate, shipping placard, control switch, lamp post`
- blocked_exits: `north`
- spatial_anomalies: `dust slips under the shutter in thin sheets`

Erreur :


```
C:\works\projects\game-liminal-raytraced-llm-  
world\documentation\generated\scene_generation_benchmark\scenes\loading_dock_dust.  
scene:8: prefab_gate requires name, pos(), size(), and gray()
```

Roof Watch

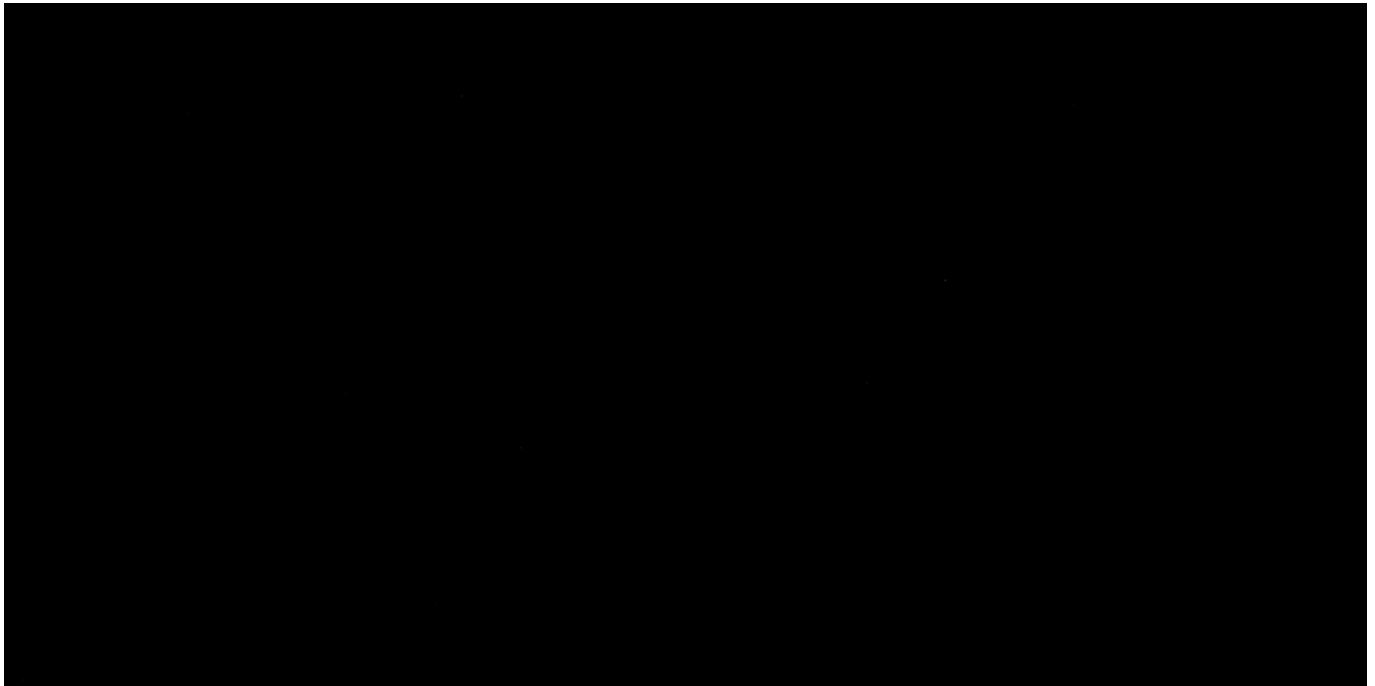
The datacenter roof becomes a parapet walk above the desert. A beacon mast, a hatch door, and a field scope hold the line against the horizon.

Status : **valid** Slug : **roof_watch_dusk** Generation : **20044.93 ms** Audit + render : **577.93 ms** Scene lines : **13** Scene chars : **1012** Triangles : **410** Materials : **35**

State : **generated/scene_generation_benchmark/states/roof_watch_dusk.session.json** Prompt : **generated/scene_generation_benchmark/prompts/roof_watch_dusk.prompt.txt** Raw response : **generated/scene_generation_benchmark/responses/roof_watch_dusk.raw.txt** Scene : **generated/scene_generation_benchmark/scenes/roof_watch_dusk.scene** Audit stdout : **generated/scene_generation_benchmark/logs/roof_watch_dusk.audit.stdout.txt** Audit stderr : **generated/scene_generation_benchmark/logs/roof_watch_dusk.audit.stderr.txt**

Spatial brief :

- location_id: **roof_watch**
- location_archetype: **roof parapet**
- time_of_day: **dusk**
- visibility_level: **clear**
- desert_state: **windy**
- interior_density: **sparse**
- alert_level: **1**
- anchors: **parapet, horizon, beacon mast, roof hatch**
- visible_objects: **roof hatch, field scope, beacon mast switch, warning sign**
- blocked_exits: **south**
- spatial_anomalies: **(none)**



Parapet Stairhead

A concrete stairhead interrupts the roof line. The landing is tight, with a latch box, a cable spool, and a narrow gate looking over the darkening desert.

Status : **valid** Slug : **stairhead_parapet** Generation : **20270.69 ms** Audit + render : **504.36 ms** Scene lines : **13** Scene chars : **1021** Triangles : **378** Materials : **34**

State : **generated/scene_generation_benchmark/states/stairhead_parapet.session.json** Prompt : **generated/scene_generation_benchmark/prompts/stairhead_parapet.prompt.txt** Raw response : **generated/scene_generation_benchmark/responses/stairhead_parapet.raw.txt** Scene : **generated/scene_generation_benchmark/scenes/stairhead_parapet.scene** Audit stdout : **generated/scene_generation_benchmark/logs/stairhead_parapet.audit.stdout.txt** Audit stderr : **generated/scene_generation_benchmark/logs/stairhead_parapet.audit.stderr.txt**

Spatial brief :

- location_id: **unknown**
- location_archetype: **roof stairhead**
- time_of_day: **dusk**
- visibility_level: **low**
- desert_state: **windy**
- interior_density: **sparse**
- alert_level: **1**
- anchors: **stairhead bulk, roof gate, parapet edge**
- visible_objects: **latch box, cable spool, roof gate handle, inspection placard**
- blocked_exits: **east, south**
- spatial_anomalies: **wind makes the gate vibrate softly**

